CRITICAL MANUFACTURING SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — the Upstream-Producer Sector

DBA-CM: Critical Manufacturing Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.4 — DRAFT

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For review by engineers, regulators, manufacturers, and policy analysts responsible for the resilience of the long-lead components and strategic materials on which every other sector depends.

SUGGESTED CITATION

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Critical Manufacturing sector. It is the parent instrument for the DBA-CM series and applies the DB → DBA → DBA-MA three-layer analytical progression to the manufacturing functions on which the physical recovery of every other sector depends: the production of long-lead critical components, primary metals and strategic materials, and precision and electronics manufacturing.

1.2 What This Document Does Not Do

This document does not supersede manufacturing, trade, or export regulation. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. It does not, at this version, develop the per-class analytical content, which the facility-class instruments named in the register carry.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The Critical Manufacturing sector is organized by function — the production of long-lead components, of primary metals and materials, and of precision electronics — and each function is placed by the downstream recovery consequence its loss produces, sub-indexed by substitutability and manufacturing lead time.

1.4 Series Placement — DBA-CM as a Standalone Sector Series

DBA-CM is a standalone sector series, a peer of the energy, water, chemical, communications, transportation, financial, healthcare, and nuclear series. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 Critical Manufacturing as the Upstream Producer — the Supply Side of RT_HVLLC

Critical Manufacturing occupies a position no other sector holds: its consequence is almost entirely downstream. A manufacturing plant's loss is rarely a large on-site event; its significance is that the component it makes cannot be replaced, and the sector that depends on that component cannot recover. This sector is therefore the supply side of a metric the corpus already uses. The energy series measures the recovery timeline for high-value long-lead components — RT_HVLLC — as a consequence of losing a large power transformer or comparable asset. That timeline is not a fixed property of the damaged asset; it is set by manufacturing capacity. RT_HVLLC is bounded by the number of plants that can build the component, their order books, and their location.

The consequence that follows is the organizing insight of this framework. An adversary who destroys a large power transformer imposes a recovery timeline measured in months; an adversary who destroys the transformer and the limited set of plants that build transformers imposes a recovery timeline measured in years, or one with no bound at all. The CIKR sector taxonomy records the underlying condition directly — large power transformers carry twelve-to-twenty-four-month lead times and limited domestic production. Critical Manufacturing capacity is the floor on every other sector's recovery, and protecting that floor is the distinctive purpose of this sector's design basis. RT_HVLLC is treated here as this sector's native governing metric rather than a borrowed one.

Section 2 — The Critical Manufacturing Sector and Its Regulatory Context

2.1 A Lightly-Regulated, Concentration-Exposed Sector

Critical Manufacturing is predominantly private and, unlike the nuclear or aviation sectors, carries no dedicated federal safety regulator. The Cybersecurity and Infrastructure Security Agency is the Sector Risk Management Agency; the Department of Commerce and its Bureau of Industry and Security govern export controls and, through the CHIPS program, semiconductor capacity; the Committee on Foreign Investment in the United States screens foreign ownership; and the Defense Production Act, particularly Title III, is the principal federal lever for surge capacity and strategic reserves. The sector's defining exposure is concentration: for the highest-consequence components, a small number of often foreign-located plants carry national supply.

2.2 Regulator-and-Mobilization Inheritance by Function

Because the applicable authority varies by function, the function-primary taxonomy is the natural structure for the sector's regulatory and mobilization terrain: long-lead components inherit the DPA Title III mobilization lever and the strategic-spares mechanisms; primary metals and materials inherit the trade and foreign-ownership regime; and precision and electronics inherit the export-control and CHIPS framework. The coordination function carries the supply-chain-risk-management and mobilization role.

2.3 The Regulatory Gap

No instrument states a design basis for manufacturing capacity as a national recovery resource. Trade policy, export controls, and the RT_HVLLC Strategic Reserve work address parts of the problem, but there is no consequence-based statement of the manufacturing capacity the nation must be demonstrated to retain in order to bound its own recovery from a large-component loss. That gap is the design-basis gap this framework begins to fill.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Substitutability Sub-Index

The sector is decomposed into three physical-function classes — long-lead critical components, primary metals and strategic materials, and precision and electronics manufacturing — and one coordination class, the manufacturing-capacity-and-coordination layer. Each is sub-indexed by substitutability and manufacturing lead time, so that the design basis load a given plant carries scales with how irreplaceable its output is.

3.2 Long-Lead Critical Components (LL)

The long-lead-component function covers the large power transformers, large generators and turbines, high- and extra-high-voltage equipment, and the large castings and forgings that underlie them. Its governing consequence is measured by RT_HVLLC in the dependent sector, and its defining property is that lead time and domestic capacity, not on-site hazard, determine its significance. It is the flagship class of the sector and the one most directly tied to the energy series.

3.3 Primary Metals & Strategic Materials (PM)

The primary-metals function covers specialty steel and alloys, the castings and forgings that feed the long-lead-component class, and the processing of strategic and rare-earth materials. Its governing consequence is upstream supply denial: the loss of a single-source material or its processing capacity propagates into every component class that depends on it, and its import concentration is a strategic vulnerability in its own right.

3.4 Precision & Electronics Manufacturing (PE)

The precision-and-electronics function covers semiconductor fabrication, the control-system components — the programmable logic controllers, remote terminal units, and specialized sensors — that instrument every other sector, and specialized electronics. Its governing consequence is loss of the compute and control substrate; it is bounded with the data-center series, and its fabrication concentration mirrors the concentration exposure of the long-lead-component class in a different technology.

3.5 CorpCo — Manufacturing-Capacity & Supply-Chain Coordination (coordination class)

The coordination function is carried by a coordination class spanning supply-chain-risk management, the Defense Production Act Title III mobilization authority, and the strategic-spares and reserve mechanisms. It is the locus at which national recovery capacity is planned and mobilized, and the locus at which the AI-Governance Design Basis is implemented for the sector — the Bright Line placement and Command-Broker obligation for the increasingly autonomous and generative-AI-driven systems in advanced manufacturing, process control, and supply-chain optimization.

† CorpCo denotes the enterprise-internal manufacturing-capacity coordination class; it is a coordination construct rather than a sector-wide operational body, with DPA Title III standing as a latent, standby government compulsion rather than a continuous operator (this is what distinguishes Critical Manufacturing from the Defense Industrial Base). The LL / PM / PE and CorpCo designators were ratified 2026-08-09.

3.6 RT_HVLLC as the Governing Metric

RT_HVLLC — the recovery timeline for high-value long-lead components, expressed in months — originated in the GenMix Optimization work and is established as a series standard in the energy framework. This framework adopts it as its own native metric, applied not to the damaged asset but to the manufacturing capacity that bounds its replacement: the design basis of a long-lead-component plant is stated in terms of the RT_HVLLC it imposes on dependent sectors if its capacity is lost, and the strategic-spares reserve is the measure that decouples RT_HVLLC from the loss of any single plant. *Long-Lead Critical Components* (DBA-CM-LL-FC1) states the manufacturing side of the metric as five parameters, build time, queue time, qualified source count, surge capacity and reserve coverage, stated by specification, and records that RT_HVLLC moves with the queue, average lead times for transformers having more than doubled between 2021 and 2024 without any attack (*Long-Lead Critical Components* §3.3, §5.3, §10.2); this framework takes the statement up at its next conformance pass.

Section 4 — The Critical Manufacturing Sector Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion is downstream recovery-consequence severity — potential for a loss of manufacturing capacity that imposes an unbounded or years-long recovery timeline on a dependent sector, or the denial of a single-source strategic material — not event likelihood. Candidate bounding events, to be developed: loss of a limited-source long-lead-component plant; loss or interdiction of a single-source strategic material or its processing; loss of a concentrated semiconductor or control-component fabrication capability; and the coordinated loss of an asset together with the capacity to replace it.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Advanced manufacturing is rapidly automating — generative design, autonomous process control, predictive-maintenance systems that can lock out equipment, and AI-driven supply-chain optimization that can reroute or halt production. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could damage a limited-source production capability or misdirect strategic-material allocation at or above the severity threshold, with the architecturally enforced authorization of a qualified Command

Broker. The inverter-firmware-trustworthiness precondition established in the energy series applies with particular force here, because this is the sector that builds the firmware-bearing components other sectors deploy. Determinism decides scope under FD-LD §4.3; it does not decide placement.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Outward View — the Recovery-Capacity Floor

The outward view is the defining one for this sector. Critical Manufacturing supplies the long-lead components and materials that every other sector's facility classes are built from — transformers and turbines to energy, pumps and treatment equipment to water, switching and fiber to communications, rolling stock and airframes to transportation. Its consequence is expressed as the RT_HVLLC it imposes on those sectors, and the forward-direction elevation runs from the dependent sector's consequence tier back to the plant: a manufacturing capability that is the sole source of a long-lead component for a Tier-2 or Tier-3 facility in another series inherits that tier, and the strategic-spares reserve is the measure that bounds the resulting obligation.

6.2 Inward View — Energy, Transportation, and Materials

The sector's reverse dependencies run to energy, whose power the energy-intensive metals and fabrication processes require; to transportation, whose specialized heavy-haul capacity moves the oversized components — a large power transformer is itself a boundary object with the DBA-TRAN conveyance class; and to the chemical and primary-materials supply that feeds production. A compound event that degrades manufacturing and the logistics that move its output extends every downstream recovery timeline simultaneously.

6.3 Concentration and Foreign Ownership

The sector shares the concentration and foreign-ownership characteristics the data-center series identified as design-basis-relevant. For the highest-consequence components, national supply rests on a small number of plants, many foreign-located, and foreign ownership creates a pre-positioning and denial vector distinct from kinetic and cyber attack. The foreign-ownership screening framework is the existing instrument; it is not calibrated to the recovery-floor consequence this framework establishes.

Section 7 — Military-Action Extension Applicability

[Skeleton.] Critical Manufacturing is the sector where the persistence and denial characteristics of military action are most consequential: the strategic objective is not the on-site event but the denial of recovery capacity. The sector-level presumption toward MA applicability is strong for the limited-source long-lead-component and strategic-material capabilities. Application remains a per-facility

determination under the mandatory two-factor test — geolocation and strategic value — with strategic value driven by irreplaceability and by the breadth of the downstream recovery its loss would deny.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-CM series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework. Each instrument adds the manufacturing-function-specific consequence chain — expressed in downstream RT_HVLLC terms — the substitutability analysis, and the recovery-capacity and strategic-reserve architecture.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-CM series. DBA-CM-LL-FC1 has issued as a working draft and the other function classes are at Anticipated status. Designators were ratified 2026-08-09 and are sub-indexed by substitutability and manufacturing lead time.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory / mobilization anchor
DBA-CM-LL-FC1	Long-lead critical components — large power transformers, large generators & turbines, HV/EHV equipment, large castings & forgings	RT_HVLLC governing metric; 12–24 mo lead; limited domestic production	Issued (working draft)	CISA (SRMA); DoD / DPA Title III
DBA-CM-PM-FC2	Primary metals & strategic materials — specialty steel & alloys, castings/forgings, rare-earth processing	single-source & import concentration	Anticipated	Commerce / BIS; CFIUS (foreign ownership)
DBA-CM-PE-FC3	Precision & electronics manufacturing — semiconductors, control-system components (PLCs, RTUs, sensors), specialized electronics	fab concentration; ties to DBA-DC / IT	Anticipated	Commerce / BIS (CHIPS); export controls
DBA-CM-CorpCo†	Manufacturing-capacity & supply-chain coordination (coordination class) — SCRM, DPA Title III, strategic-spares reserve	recovery-capacity mobilization	Anticipated (architecture)	DHS / CISA; DoD (DPA); NARUC (rate recovery)

Designators were ratified 2026-08-09 at v0.2. Facility-class numbers are sequential across the series; CorpCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Each class is sub-indexed by substitutability / manufacturing lead time.

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-CM sector framework. It establishes the standalone-series placement, the upstream-producer framing and the manufacturing-capacity-as-recovery-floor insight, the function-primary decomposition (long-lead components, primary metals, precision and electronics, and the CorpCo capacity-and-coordination class) with a substitutability sub-index, the adoption of RT_HVLLC as the sector's native governing metric, and the concentration and foreign-ownership exposures shared with the data-center series. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: develop the long-lead-component instrument (DBA-CM-LL-FC1) as the Phase-A instrument on the RT_HVLLC Strategic Reserve basis; and reconcile the reversed-direction recovery-floor elevation with the energy series registers. Note that the CIKR sector mindmap carries no Critical Manufacturing sector, only a supply-chain-equipment fragment — a documented gap for the mindmap reconciliation. At v0.4, the Phase-A instrument named above has issued: DBA-CM-LL-FC1 (*Long-Lead Critical Components*), the series' first facility class, physical, and the three reference events it proposed are conferred at Section 9.1. It develops RT_HVLLC as this sector's native metric on its manufacturing side and the candidate bounding events of Section 4.1 for the long-lead class. What it notes to this framework is taken up at the framework's next conformance pass: the five parameters, the event categories it develops ahead of Section 4, the three tiers stated by specification, its declared residuals and the consequence-basis conditionality check (*Long-Lead Critical Components* §10.10). It notes to the energy series, in Project 8, that RT_HVLLC should carry a surge figure beside each normal-condition value, and that the strategic-reserve paper's description of the Spare Transformer Equipment Program should be corrected (*Long-Lead Critical Components* §10.2, §10.4). The cover of v0.3 was dated August 2026 although that version was made in October 2026; v0.4 is dated October 2026. Further open item: the three reference-event instruments of Section 9.1.

9.1 Conferred Reference Events

On the author's decision of 14 October 2026, this framework confers a reference-event designator on each of the three events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the three instruments is yet written. The designators are numbered in the order the proposing class lists its events; the number records conferral, not issue. Each designator is sector-primary and single.

From *Long-Lead Critical Components*, three events are conferred: DBA-CM-RE-1, the denial of large-transformer replacement in Ukraine, the closure of Zaporozhtransformator by the war and the campaign against the power system, from 2022; DBA-CM-RE-2, the shortage of transformers in the United States, from 2021; and DBA-CM-RE-3, the cyberattack on a German steel mill reported by the Federal Office for Information Security in 2014, read at the boundary

with *Primary Metals and Strategic Materials* (DBA-CM-PM-FC2). The Ukrainian and shortage events are continuing conditions, and their instruments bound them. The fire at the Renesas Naka semiconductor plant (2021) and the attack on a Toyota supplier (2022), which the plan for closing the seven sectors named for this sector, are not conferred here; they are left to *Precision and Electronics Manufacturing* (DBA-CM-PE-FC3). The three instruments are owed, and they are Project 11's to write. *Long-Lead Critical Components* states the three as proposed at its v0.1 and conforms to the conferral at its next revision.

This framework is still a skeleton at Sections 4, 5 and 7. The three instruments read their events back through the design basis of *Long-Lead Critical Components*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); GenMix Optimization Whitepaper (origin of RT_HVLLC); RT_HVLLC Strategic Reserve WP (Option D, DPA Title III, NARUC); DBA-EN (RT_HVLLC series standard; downstream component dependency); DBA-DC (concentration and foreign-ownership characteristics); DBA-TRAN (heavy-haul conveyance boundary). Regulatory and mobilization: CISA (SRMA); Commerce / BIS and CHIPS; CFIUS; DoD / DPA Title III. Series instruments: DBA-CM-LL-FC1 (*Long-Lead Critical Components*), the series' first facility class; DBA-CM-RE-1 to DBA-CM-RE-3, the conferred reference events of Section 9.1, their instruments owed to Project 11.

Revision History

0.4 October 2026 — T. Roxey. Section 9.1 is added and confers DBA-CM-RE-1 to DBA-CM-RE-3 on the three reference events that DBA-CM-LL-FC1 (Long-Lead Critical Components) proposed, on the author's decision of 14 October 2026: the denial of transformer replacement in Ukraine (from 2022), the United States transformer shortage (from 2021) and the German steel-mill attack (2014). Renesas and Kojima are left to Precision and Electronics. Each designator commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's. The register at Section 8.2 enters the class as issued. Section 3.6 records the class's statement of the manufacturing side of RT_HVLLC. Section 9 records the class's issue, what it notes to this framework and to the energy series, and the correction of the cover date. The cover now reads October 2026; v0.3's cover read August 2026 although v0.3 was made in October. References add the class and the three designators. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and

states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified; coordination class resolved to CorpCo (enterprise-internal; DPA Title III latent); LL / PM / PE confirmed; RT_HVLLC native governing metric confirmed.

0.1 August 2026 — T. Roxey. Initial skeleton. Established DBA-CM as a standalone sector series with the upstream-producer framing: the sector's consequence is downstream, and manufacturing capacity is the floor on every other sector's recovery (the supply side of RT_HVLLC). Checked against the CIKR mindmap, which carries the underlying condition (LPT 12–24 month lead, limited domestic production, manufacturing-capacity constraints) only as a supply-chain fragment, not a sector — a documented gap for the reconciliation. Adopted the function-primary decomposition (Long-Lead Components / Primary Metals / Precision & Electronics / CorpCo) with a substitutability sub-index; adopted RT_HVLLC as the sector's native governing metric; carried the concentration and foreign-ownership exposures from the data-center series. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.